

DS 413 : Introduction to Statistical Learning

Course Code : DS 413

Course Name : Introduction to Statistical Learning

L-T-P-C : 3-1-0-4

Intended for : Discipline core for B. Tech. Data Science and Engineering, Discipline elective for B. Tech. Computer Science and Engineering, Electrical Engineering, Free elective for other B. Tech. disciplines

Prerequisite : DS201 - Data Handling and visualization or equivalent, DS303 – Statistical Foundations of Data Science or equivalent, DS402 – Matrix Computations for Data Science; From 2022 UG Batches

Mutual Exclusion:

Approval: 55th BoA

Course Contents

- **Introduction to learning from data:** Introduction to supervised learning and unsupervised learning. (2 Lectures)
- **Supervised learning: Regression:** Linear regression models and least squares, Shrinkage methods: ridge regression and the LASSO. (6 Lectures)
- **Supervised learning: Classification:** Logistic regression, nearest neighbour's method, Bayes classifier with unimodal and multimodal density - maximum likelihood estimation, expectation-maximization (EM) algorithm; decision trees, support vector machines (SVMs), basics of neural networks. (8 Lectures)
- **Model Assessment and Selection:** Bias, variance and model complexity, The Bayesian approach, AIC and BIC, cross-validation, bootstrap methods, hypothesis testing, confidence intervals, significance testing. (6 Lectures)
- **Unsupervised learning:** Introduction to association rules, clustering, and dimension reduction. (6 Lectures)

Lab Exercises:

Lab to be conducted on a 2-hour slot. It will be conducted in tandem with the theory course so the topics for problems given in the lab are already initiated in the theory class. The topics taught in the theory course should be appropriately be sequenced for synchronization with the laboratory.

Textbooks:

1. Hastie, T., Tibshirani, R. and Friedman, J., **The Elements of Statistical Learning: Data Mining, Inference, and Prediction**, 2nd Edition, 12th Reprint, Springer, 2017.

References:

1. Duda, R. O., Hart, P. E. and Stork, D. G., **Pattern Classification**, John Wiley, 2001.
2. Bishop, C. M., **Pattern Recognition and Machine Learning**, Springer, 2006.
3. Theodoridis, S. and Koutroumbas, K., **Pattern Recognition**, Academic Press, 2009.